

XITAI, China

WMO: 588530

Lat: 27.008N

Lon: 120.695E

Elev: 107

StdP: 100.05

Time Zone: 8.00 (E08)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.0	3.8	-7.1	2.1	5.6	-5.0	2.5	5.7	18.3	5.7	17.2	6.1	10.9	20	0.716

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	29.9	26.4	29.2	26.3	28.5	26.0	27.0	28.8	26.7	28.3	26.3	27.8	4.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.3	28.0	26.2	21.8	27.7	25.9	21.5	27.3	86.0	29.2	84.4	28.5	83.0	28.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.4	15.1	1.5	31.9	1.7	1.5	0.3	33.0	-0.7	33.9	-1.7	34.7	-2.9	35.8		
(5)			-1.2	27.8	1.2	0.4	-2.1	28.1	-2.8	28.3	-3.5	28.5	-4.4	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	8.5	8.0	9.9	14.2	19.3	23.3	26.1	27.0	24.9	20.7	16.5	11.1		
	DBStd	7.15	2.61	2.61	2.54	2.67	2.49	1.93	1.43	1.17	2.08	2.33	2.65	3.25		
	HDD10.0	189	61	66	32	2	0	0	0	0	0	0	1	27		
	HDD18.3	1295	304	288	260	127	20	0	0	0	0	6	66	225		
	CDD10.0	2933	15	11	31	128	288	398	498	528	448	333	195	61		
	CDD18.3	998	0	0	0	2	49	149	240	269	198	81	9	0		
	CDH23.3	6532	0	0	0	0	25	572	1961	2548	1298	126	2	0		
	CDH26.7	877	0	0	0	0	1	22	228	471	148	8	0	0		
(14) Wind	WSAvg	8.1	9.0	9.0	8.1	6.4	6.7	7.5	8.4	7.0	8.4	9.2	9.4	8.6		
(15) Precipitation	PrecAvg	1099	50	74	119	127	155	180	46	98	116	51	49	33		
	PrecMax	1496	169	142	294	328	414	392	153	302	336	192	156	94		
	PrecMin	465	2	15	19	8	29	31	5	8	7	0	0	0		
	PrecStd	251	44	32	77	74	90	97	46	67	86	46	39	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	16.5	18.0	21.7	25.2	28.2	30.5	31.4	30.0	27.8	22.8	18.6		
		MCWB	12.9	13.7	16.5	19.8	22.4	26.0	27.0	26.4	26.4	25.1	20.5	15.3		
	2%	DB	14.2	14.5	16.0	20.0	24.0	27.2	29.4	30.3	29.1	25.6	21.2	16.9		
		MCWB	12.0	12.9	14.6	18.8	22.3	25.5	26.6	26.3	26.0	23.9	19.4	14.6		
	5%	DB	13.0	13.0	14.7	18.9	23.2	26.6	28.7	29.6	28.2	24.6	20.5	16.0		
		MCWB	11.4	11.3	13.7	18.1	21.9	25.2	26.4	26.2	25.6	22.8	18.8	13.9		
	10%	DB	11.9	11.6	13.6	17.9	22.5	25.9	28.0	28.9	27.5	23.6	19.7	15.2		
		MCWB	10.4	10.2	12.5	16.7	21.4	24.8	26.0	26.1	25.3	21.4	18.1	13.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	14.9	17.0	20.4	23.7	26.5	27.6	27.5	27.1	25.5	21.0	16.4		
		MCDB	15.2	15.7	17.7	21.2	24.4	27.5	29.5	29.6	28.7	27.0	21.8	17.6		
	2%	WB	12.7	13.3	15.2	19.2	23.0	25.8	27.0	27.0	26.6	24.6	20.1	15.4		
		MCDB	13.7	14.2	15.6	19.5	23.6	26.7	28.6	29.1	28.1	25.5	20.9	16.6		
	5%	WB	11.7	11.7	14.2	18.2	22.3	25.4	26.6	26.7	26.1	23.6	19.4	14.4		
		MCDB	12.7	12.5	14.7	18.8	22.9	26.2	28.1	28.5	27.6	24.4	20.2	15.7		
	10%	WB	10.6	10.5	12.7	17.2	21.7	25.0	26.2	26.4	25.7	22.1	18.7	13.4		
		MCDB	11.6	11.4	13.4	17.8	22.2	25.7	27.5	28.0	27.1	23.2	19.6	15.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.8	2.6	2.8	3.2	3.0	2.7	2.9	3.3	2.8	2.5	2.7	3.0		
		MCDBR	5.2	5.2	4.7	4.6	4.2	3.3	4.0	4.4	3.7	3.6	3.6	4.2		
	5% WB	MCWBR	3.6	4.0	3.9	3.8	2.9	1.8	2.0	2.0	2.1	2.4	2.9	3.5		
		MCWBR	4.7	4.1	4.7	4.4	3.5	2.9	3.6	3.9	3.4	3.4	3.5	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.575	0.625	0.684	0.685	0.620	0.580	0.537	0.565	0.556	0.561	0.559	0.547		
	taud		1.754	1.689	1.602	1.660	1.841	1.971	2.101	2.012	2.001	1.899	1.860	1.829		
	Ebn at Noon		650	658	651	670	716	740	772	749	741	703	662	652		
	Edh at Noon		198	228	261	253	211	184	162	176	173	181	177	177		
(44) All-Sky Solar Radiation	RadAvg		2.17	2.47	3.28	3.76	4.17	4.49	6.13	5.49	4.50	3.70	2.48	2.27		
	RadStd		0.49	0.57	0.34	0.45	0.65	0.52	0.48	0.45	0.44	0.62	0.38	0.41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page